

FlexPoint Pipeline Intelligence

CSV Data Dictionary for Gallus Insights

This doc is a short bridge between the 22 CSVs in this folder and the FlexPoint dashboard (01_dashboard_reference.html). For each CSV you get: what it contains, which dashboard visual it feeds, and a one-line recipe for recreating that visual in ThoughtSpot. Open the HTML side-by-side with this PDF to see the target while you read.

How to use the handoff bundle

1. Open 01_dashboard_reference.html in any browser - this is the visual target.
2. Skim this PDF's CSV index on the next page.
3. For each chart in ThoughtSpot, find the matching CSV entry, follow the recipe.
4. loans.csv is the master fact table - everything else joins to it on LoanGuid.
5. If you want to regenerate on fresh data, see 03_model_code/RUNBOOK.md.

Naming conventions across CSVs

Same concept = same column name across every CSV. 'Branch Channel', 'Product Type', 'LoanAmount', 'LoanGuid' use the raw sectG spelling (spaces, capitalization preserved) so they join cleanly. Counts use 'loan_count'. Probabilities use 'avg_probability'. \$ totals use 'total_value' (overall) or 'expected_value' (probability-weighted).

Snapshot date: 2025-12-15 | Model: GradientBoosting v3 (5.8% MAPE backtest)

CSV Index - What Each File Is For

CSV	Dashboard Tab	One-line purpose
loans.csv *	Overview	Master per-loan table.
already_funded.csv	Overview	Loans that funded earlier this month before the snapshot date - the 'already booked' part of the monthly total.
summary_kpis.csv	Overview	Single-row snapshot of every top-of-dashboard KPI value (pipeline counts, \$ totals, pull-through, elimination breakdown).
stage_funnel.csv	Overview	Loan count and dollar volume at each pipeline stage.
channel_split.csv	Overview	Wholesale vs Retail breakdown of pipeline volume.
product_breakdown.csv	Overview	Pipeline volume by Product Type.
pull_through_monthly.csv	Trends	Historical monthly pull-through rate by product (% of loans opened in a month that eventually funded).
cycle_times_by_product.csv	Trends	Distribution stats (median, mean, percentiles) of total funded cycle time by product, with SLA-breach rate.
cycle_times_per_loan.csv	Trends	Per-loan total cycle time - the raw data behind the cycle-time charts.
backtest_monthly.csv	Overview	Model accuracy vs actuals across 24 backtested months (Jan 2024 - Dec 2025).
revenue_at_risk_buckets.csv	Revenue at Risk	Dollars at risk bucketed by failure archetype (lock expired, stuck, etc).
moneyball_quadrant_summary.csv	Revenue at Risk	Count and dollar value of loans in each Moneyball quadrant (easy_win / quick_fix / stretch / long_shot).
bottleneck_heatmap.csv	Pipeline Health	Stage transition times by product with industry benchmarks - the Pipeline Health heatmap.
stage_conversion_rates.csv	Pipeline Health	What % of loans that reach a stage actually fund (per product).
current_bottlenecks.csv	Pipeline Health	Where live loans are piling up right now (count of loans stuck per stage, with \$ stuck).
velocity_distribution.csv	Pipeline Health	Distribution of pipeline velocity bands (Fast / Normal / Slow / Stalled) by product.
velocity_by_stage.csv	Pipeline Health	Median velocity (and IQR) for live loans at each pipeline stage.
what_if_scenarios.csv	What-If	Impact of 4 operational levers (lock improvements, channel-mix shifts, etc).
scorecards.csv	Scorecards	Balanced composite score (0-100) per product / channel with five sub-scores.
optimization_recommendations.csv	Overview	Top ranked actions with estimated dollar impact (the 'tell me what to do' list).
industry_benchmarks_transitions.csv	Pipeline Health	ICE Mortgage Technology 2024 industry benchmarks for stage transitions by product.
feature_importance.csv	Not on dashboard	Model feature importance ranking - which features drive the predictions.

* = master fact table. All per-loan drill-down queries start here.

CSV Details

One entry per CSV. Open the HTML dashboard alongside this - the tab + chart name in each entry points to the visual you're trying to reproduce.

loans.csv [MASTER TABLE]

What it is: Master per-loan table. Every other CSV aggregates this one or references it by LoanGuid.

Grain: One row per active pipeline loan at snapshot_date (2025-12-15).

Dashboard chart(s) it feeds:

[Overview]	KPI tiles - count live loans, sum expected_value
[Overview]	Stage funnel - group by current_stage
[Watch List]	Loan table - filter is_at_risk = TRUE
[Revenue at Risk]	Moneyball bubble - moneyball_difficulty x ml_probability, size=LoanAmount
[Revenue at Risk]	Top recovery table - filter recovery_rank IS NOT NULL ORDER BY recovery_rank
[Any tab]	Drill-down target - join other CSVs back via LoanGuid

ThoughtSpot recipe:

Load as the base worksheet. Every other worksheet joins back to this on LoanGuid. Most dashboard charts are this table filtered + grouped.

Key columns:

LoanGuid - join key (used by every other CSV)

LoanAmount, ml_probability, expected_value, base_probability - core revenue metrics

current_stage, stage_rank, days_at_stage - where the loan is and how long it's been there

Product Type, Branch Channel, Loan Purpose - primary slicing dimensions

status, is_at_risk, elimination_reason, failure_archetype, risk_reasons - live/dead + risk filter flags

recommended_action, counterfactual_probability, probability_delta, expected_value_uplift - counterfactual recommendation

moneyball_quadrant, moneyball_difficulty, moneyball_is_movable, moneyball_cf_quadrant - Moneyball chart axes

recovery_rank, recovery_gap, momentum_rank - ranked subsets - filter NOT NULL for top-N tables

velocity, velocity_band - pipeline velocity classification

f_* (31 columns) - engineered features fed to the ML model (audit / debug)

Loan Number, Loan Status, IsFunded, DecisionCreditScore, LTV, CLTV, NoteRate, Lock Period (days), ... - raw passthrough columns from sectG (~150 fields)

already_funded.csv

What it is:	Loans that funded earlier this month before the snapshot date - the 'already booked' part of the monthly total.
Grain:	One row per loan funded between Dec 1 and Dec 15, 2025.
Dashboard chart(s) it feeds:	
[Overview]	Monthly total KPI - \$68M already funded + \$32.4M projected = ~\$100M
[Overview]	Channel split for realized fundings
ThoughtSpot recipe:	
Sum LoanAmount. Display next to sum(expected_value) from loans.csv to get the full monthly projection.	
Key columns:	
snapshot_date, LoanGuid, Funded D - identity + fund date	
LoanAmount - amount to sum	
Product Type, Branch Channel, Loan Purpose - breakdown dimensions	

summary_kpis.csv

What it is:	Single-row snapshot of every top-of-dashboard KPI value (pipeline counts, \$ totals, pull-through, elimination breakdown).
Grain:	ONE row total. Wide format - each column is a different KPI value.
Dashboard chart(s) it feeds:	
[Overview]	KPI tile row (top of dashboard)
ThoughtSpot recipe:	
Use as a one-row lookup table. Pin individual columns as KPI tiles - no aggregation needed.	
Key columns:	
snapshot_date, month, model_used - context	
total_pipeline_loans, total_pipeline_value, live_pipeline_loans, live_pipeline_value, dead_pipeline_loans, dead_pipeline_value - pipeline counts + \$	
already_funded_loans, already_funded_value, projected_total - monthly projection	
overall_pull_through, median_cycle_days - headline metrics	
elimination_total, elimination_count, elimination_pct, elim_opened_stale, elim_underwriting_unlocked_stale, elim_approved_expired_lock, elim_submitted_unlocked_stale, elim_application_stale - elimination filter breakdown	

stage_funnel.csv

What it is: Loan count and dollar volume at each pipeline stage.

Grain: One row per pipeline stage.

Dashboard chart(s) it feeds:

[\[Overview\]](#) Pipeline funnel chart

ThoughtSpot recipe:

Column or funnel chart. X = stage (sorted by rank). Y = loan_count or total_value. Add live_loans / live_value as overlay.

Key columns:

stage, rank - X axis (rank = stage ordering)

loan_count, total_value - Y axis options

live_loans, live_value, avg_probability - live-pipeline subset

snapshot_date - as-of date

channel_split.csv

What it is: Wholesale vs Retail breakdown of pipeline volume.

Grain: One row per Branch Channel.

Dashboard chart(s) it feeds:

[\[Overview\]](#) Channel split donut / bar

ThoughtSpot recipe:

Donut or bar. Category = Branch Channel. Value = loan_count or projected_value.

Key columns:

Branch Channel - category

loan_count, total_value, projected_value, avg_probability - metrics

live_loans, live_value - live-only subset

snapshot_date - as-of date

product_breakdown.csv

What it is: Pipeline volume by Product Type.

Grain: One row per Product Type.

Dashboard chart(s) it feeds:

[\[Overview\]](#) Product mix bar chart

ThoughtSpot recipe:

Bar: X = Product Type, Y = loan_count or projected_value.

Key columns:

Product Type - category

loan_count, total_value, projected_value, avg_probability - metrics

live_loans, live_value - live-only subset

snapshot_date - as-of date

pull_through_monthly.csv

What it is: Historical monthly pull-through rate by product (% of loans opened in a month that eventually funded).

Grain: One row per (month x Product Type).

Dashboard chart(s) it feeds:

[\[Trends\]](#) Monthly pull-through line chart

ThoughtSpot recipe:

Line. X = month. Y = pull_through_rate. Color by Product Type. Filter Product Type='Overall' for the headline line.

Key columns:

month, Product Type - axes (Product Type = 'Overall' = aggregate line)

pull_through_rate, funded_count, total_count - rate + counts

cycle_times_by_product.csv

What it is: Distribution stats (median, mean, percentiles) of total funded cycle time by product, with SLA-breach rate.

Grain: One row per Product Type.

Dashboard chart(s) it feeds:

[\[Trends\]](#) Cycle time distributions with IQR bands

ThoughtSpot recipe:

Box-style bar. X = Product Type. Y = median. Error bars = p25 to p75 (or p10 to p90).

Key columns:

Product Type, loan_count - category + sample size

median, mean, std, p10, p25, p75, p90 - distribution

above_sla, above_sla_pct, sla_threshold_days - SLA-breach stats

cycle_times_per_loan.csv

What it is: Per-loan total cycle time - the raw data behind the cycle-time charts.

Grain: One row per funded loan.

Dashboard chart(s) it feeds:

[\[Trends\]](#) Histogram / scatter - raw distribution of cycle times

ThoughtSpot recipe:

Histogram of cycle_days. Or scatter: X = Funded D, Y = cycle_days, color = Product Type.

Key columns:

LoanGuid - join to loans.csv

Product Type, Branch Channel, Loan Purpose - slicing dimensions

LoanAmount, Funded D, cycle_days - main metrics

backtest_monthly.csv

What it is: Model accuracy vs actuals across 24 backtested months (Jan 2024 - Dec 2025).

Grain: One row per (month x snapshot_day x method x training_mode).

Dashboard chart(s) it feeds:

[\[Overview\]](#) Backtest accuracy chart - projected vs actual

ThoughtSpot recipe:

Two-line chart. Filter method='ML' AND training_mode='rolling' AND snapshot_day=15. X = year-month. Y1 = projected, Y2 = actual. Secondary axis = error_pct.

Key columns:

year, month, snapshot_day, method, training_mode - filter to one series

already_funded, projected_pipeline, projected, actual - \$ values

error_pct, direction - accuracy

revenue_at_risk_buckets.csv

What it is: Dollars at risk bucketed by failure archetype (lock expired, stuck, etc.).

Grain: One row per risk bucket.

Dashboard chart(s) it feeds:

[\[Revenue at Risk\]](#) Risk bucket bar chart (\$145M total)

ThoughtSpot recipe:

Horizontal bar. Category = label. Value = value_at_risk. Sort descending. Tooltip = description + action.

Key columns:

id, label, description, action - bucket identity + recommended action

loan_count, total_value, expected_value, value_at_risk, avg_probability - metrics (value_at_risk = bar height)

_total_at_risk, _total_recovery_potential, _live_pipeline_value - row-constant grand totals

moneyball_quadrant_summary.csv

What it is: Count and dollar value of loans in each Moneyball quadrant (easy_win / quick_fix / stretch / long_shot).

Grain: One row per quadrant.

Dashboard chart(s) it feeds:

[\[Revenue at Risk\]](#) Moneyball quadrant legend / summary tiles

ThoughtSpot recipe:

4 KPI tiles, one per quadrant. Value = loan_count or total_value. Pair with the bubble chart from loans.csv.

Key columns:

quadrant - category

loan_count, total_value, total_expected_value, avg_probability - per-quadrant metrics

movable_count, total_loans_all_quadrants, snapshot_date - row-constant grand totals + as-of

bottleneck_heatmap.csv

What it is: Stage transition times by product with industry benchmarks - the Pipeline Health heatmap.

Grain: One row per (transition x Product Type) cell.

Dashboard chart(s) it feeds:

[\[Pipeline Health\]](#) Main heatmap: transitions x products, colored by vs_benchmark

ThoughtSpot recipe:

Heatmap. Rows = transition. Columns = Product Type. Color cell = vs_benchmark (green=faster than industry, red=slower).

Tooltip: median_days + p25/p75 + benchmark_median.

Key columns:

transition, Product Type - heatmap axes

median_days, p25_days, p75_days, std_days - FlexPoint cell + IQR

benchmark_median, benchmark_fast, benchmark_slow, vs_benchmark - industry comparison (vs_benchmark = days faster/slower)

stage_conversion_rates.csv

What it is: What % of loans that reach a stage actually fund (per product).

Grain: One row per (Product Type x stage).

Dashboard chart(s) it feeds:

[\[Pipeline Health\]](#) Conversion funnel

ThoughtSpot recipe:

Funnel chart. Filter Product Type='Overall'. X = stage (ordered). Y = conversion_rate.

Key columns:

Product Type, stage - axes

reached_count, funded_count, conversion_rate - metrics

current_bottlenecks.csv

What it is: Where live loans are piling up right now (count of loans stuck per stage, with \$ stuck).

Grain: One row per (Product Type x stage). Top-ranked stages have rank populated.

Dashboard chart(s) it feeds:

[\[Pipeline Health\]](#) Current bottleneck pileup bar chart

ThoughtSpot recipe:

Bar. Filter Product Type='Overall'. X = stage. Y = loan_count. Color by pct_over_30d.

Key columns:

Product Type, stage, rank - axes (rank = bottleneck-severity rank)

loan_count, total_value - size of pileup

avg_days_at_stage, median_days_at_stage, p25_days_at_stage, p75_days_at_stage - how long they have been stuck

avg_probability, pct_over_30d - severity

snapshot_date - as-of

velocity_distribution.csv

What it is: Distribution of pipeline velocity bands (Fast / Normal / Slow / Stalled) by product.

Grain: One row per (Product Type x band).

Dashboard chart(s) it feeds:

[\[Pipeline Health\]](#) Velocity distribution donut

ThoughtSpot recipe:

Donut. Filter Product Type='Overall'. Category = band. Value = loan_count or expected_value.

Key columns:

Product Type, band - axes

loan_count, total_value, expected_value, avg_probability - metrics

snapshot_date - as-of

velocity_by_stage.csv

What it is: Median velocity (and IQR) for live loans at each pipeline stage.

Grain: One row per stage.

Dashboard chart(s) it feeds:

[\[Pipeline Health\]](#) Velocity by stage bar chart with IQR

ThoughtSpot recipe:

Bar. X = stage. Y = median_velocity. Error bars = p25_velocity to p75_velocity. Annotate avg_days_at_stage.

Key columns:

stage, loan_count - axes + sample size

avg_velocity, median_velocity, p25_velocity, p75_velocity - velocity distribution

avg_days_at_stage, avg_probability, pct_stalled - severity

what_if_scenarios.csv

What it is: Impact of 4 operational levers (lock improvements, channel-mix shifts, etc.) on projected fundings.

Grain: One row per scenario.

Dashboard chart(s) it feeds:

[\[What-If\]](#) Scenario cards with raw vs overlap-adjusted values

ThoughtSpot recipe:

Card layout. Each card shows lever, description, current_state -> target_state, delta (\$), pct_improvement, confidence. Footer = caveats.

Key columns:

id, lever, description - scenario identity

current_state, target_state, current_value, improved_value - scenario definition

delta, pct_improvement, affected_loans, affected_value - raw impact

methodology, confidence, confidence_note, caveats - trustworthiness

totals* (8 columns) - row-constant grand totals (raw + overlap-adjusted upside)

scorecards.csv

What it is: Balanced composite score (0-100) per product / channel with five sub-scores.

Grain: One row per scorecard entity (product or channel).

Dashboard chart(s) it feeds:

[\[Scorecards\]](#) Product / channel scorecard grid

ThoughtSpot recipe:

Sortable table. Sort by rank (or composite_score desc). Color composite_score on a gradient. Tier = Elite/Strong/Average/Weak. Filter by dimension to split product vs channel boards.

Key columns:

name, dimension, rank, tier, composite_score - ranking

sub_pull_through, sub_cycle_time, sub_revenue_efficiency, sub_trend, sub_pipeline_probability - 5 weighted sub-scores

pull_through_rate, pt_recent_3m, pt_prior_3m, pt_trend_delta, pt_trend, median_cycle_days, avg_loan_amount,

funded_volume_6m, pipeline_volume_6m, revenue_efficiency, current_active_loans, current_projected_value,

avg_pipeline_probability, efficiency_score - underlying raw values

industry_benchmark_pt, industry_benchmark_cycle, benchmark_note - industry comparison

optimization_recommendations.csv

What it is: Top ranked actions with estimated dollar impact (the 'tell me what to do' list).

Grain: One row per recommendation.

Dashboard chart(s) it feeds:

[\[Overview\]](#) Recommendations panel at top of dashboard

ThoughtSpot recipe:

Ordered list / table. Sort by priority. Display: priority, title, description, estimated_impact, urgency, confidence_caveat.

Key columns:

priority, title, description - the action

estimated_impact, effort, loan_count, category, urgency - impact + effort

confidence_caveat - trust note

_total_estimated_impact_all, snapshot_date - row-constant grand total + as-of

industry_benchmarks_transitions.csv

What it is: ICE Mortgage Technology 2024 industry benchmarks for stage transitions by product.

Grain: One row per (transition x Product Type).

Dashboard chart(s) it feeds:

[\[Pipeline Health\]](#) Reference data for heatmap overlay

ThoughtSpot recipe:

Lookup table. Already pre-joined into bottleneck_heatmap.csv as benchmark_median/fast/slow. Show standalone as a small reference table on the methodology page.

Key columns:

transition, Product Type - lookup keys
benchmark_fast, benchmark_median, benchmark_slow - industry stats
source - citation

feature_importance.csv

What it is: Model feature importance ranking - which features drive the predictions.

Grain: One row per feature.

Dashboard chart(s) it feeds:

[\[Not on dashboard\]](#) Reference only - for model audit / explainability

ThoughtSpot recipe:

Horizontal bar chart sorted by importance desc. Top 10 typically shown.

Key columns:

feature, importance

Quick reference: where does each dashboard tab's data come from?

Overview	summary_kpis, stage_funnel, channel_split, product_breakdown, backtest_monthly, optimization_recommendations, already_funded + loans
Watch List	loans (filter is_at_risk = TRUE)
Revenue at Risk	revenue_at_risk_buckets, moneyball_quadrant_summary, loans (recovery_rank, moneyball_*)
Pipeline Health	bottleneck_heatmap, stage_conversion_rates, current_bottlenecks, velocity_distribution, velocity_by_stage, industry_benchmarks_transitions
Trends	pull_through_monthly, cycle_times_by_product, cycle_times_per_loan
What-If	what_if_scenarios
Scorecards	scorecards

Questions while building?

Reach out to Ajer (ajersher61@gmail.com). The most common questions are about the model column definitions (the f_* features in loans.csv) and the counterfactual recommendation logic - both are documented in 03_model_code/src/feature_engineering_v3.py and scorer.py if you want to dig in.